Tribunal — Product Requirements Document (PRD)

1. Problem Statement

Single-model LLM responses to complex, high-stakes decisions suffer from a structural weakness: one model reasoning alone has no adversarial pressure-testing built in. It can miss risks, overstate confidence, or produce plausible-sounding but shallow analysis. Existing multi-agent frameworks (AutoGen, CrewAI-style group chats) attempt to fix this with free-form agent conversation, but this often produces the opposite failure — agents converge too easily (consensus bias/sycophancy), context bloats across turns, and errors compound rather than get caught.

2. Product Goal

Build a deterministic, structured pipeline that forces genuine adversarial reasoning on a decision question — not a chatbot, not a group discussion — and produces a clear, evidence-aware output a human can act on.

3. Core Principle

Structure disagreement deliberately. Don't let it emerge from free conversation (which drifts toward agreement); enforce it through architecture:

- Agents reason independently before seeing each other's work (no anchoring)
- A fresh, uninvolved agent does the attacking (no self-contamination)
- Claims are stripped of identity before cross-examination (no bias toward "my side")
- Severity and confidence are rated by agents, not presupposed by their role

4. v0 Scope (current, in progress)

In scope:

- Single decision domain (business/investment decisions — e.g. "should we acquire company X")
- Proponent agent — builds bull case, structured JSON output — **built, working**
- Antagonist agent — builds bear case, structured JSON output — **next**
- Claim extraction — deterministic code, strips agent identity
- Examiner agent — fresh context, cross-examines stripped claims, rates attack severity (minor/moderate/fatal)

- Output: side-by-side bull case, bear case, flagged weaknesses — no final verdict, human reads and decides

Explicitly out of v0:

- Fact-checking / web verification
- Judge/synthesis agent producing a final Decision Matrix
- Human tie-break checkpoint
- Multiple decision domains
- Dynamic agent selection or workflow routing
- Streaming (SSE) — simple loading state is enough for v0

5. v1 — Next Scaling Step (after v0 is proven)

- Add Verification Node: batch empirical claims, check against web search (Tavily), tag VERIFIED/DISPUTED/UNSUBSTANTIATED
- Add Judge agent: synthesizes surviving claims into a Decision Matrix (summary, trade-offs, fatal flaws)
- Add computed deadlock detection: rule-based (not LLM-judged) trigger for human tie-break when a fatal, disputed claim blocks resolution
- Real-time SSE streaming of pipeline progress

6. v2+ — Bigger Vision (future, not committed scope)

- Multiple decision domains beyond business/investment (research, hiring, technical architecture decisions)
- Domain-specific agent tuning per decision type
- Session history / past-decision comparison
- Possible dynamic agent count for unusually complex decisions — **only after v0/v1 prove the fixed pipeline adds real value**; this is explicitly the highest-risk, most speculative expansion and should not be pulled forward

7. Success Criteria for v0

- Pipeline runs end-to-end without manual intervention
- Examiner's attacks are judged (by manual review) as substantive, not decorative, on at least 70% of a 10-question test set
- At least one clear example where the pipeline surfaces a flaw a single-model answer to the same question misses

8. Non-Goals

- Not a general-purpose chatbot

- Not trying to replace human judgment — output informs a decision, doesn't make it
- Not optimizing for speed/cost at the expense of adversarial rigor in v0 (that tradeoff can be revisited once the mechanism is proven)